Venkata Lakshmi Parimala Pasupuleti

Software Engineer · AI Engineer · Python · 9yr Production · LangGraph · RAG Pipelines · LLM Evaluation · AWS · Docker · Agentic AI

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github.com/venkatalakshmiparimala | Virginia, USA · Open to Relocation

PROFESSIONAL SUMMARY

AI engineer with 9+ years of production software engineering experience who builds and ships agentic AI systems end-to-end - not demos, production. AgentProbe is an open-source multi-provider LLM evaluation platform I built from scratch: routes inference across Anthropic, OpenAI, and Ollama via clean API abstractions, runs a 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), and gates quality regressions in CI before anything ships. Leasel is a three-surface production AI platform - 8-agent LangGraph investigation pipeline with HITL escalation, RAG over complex regulatory documents with 100% citation correctness, and a deterministic guardrail layer that ensures every regulated decision goes through a human. At Entrata, I owned compliance-sensitive production systems for 9+ years (AWS, Docker, RabbitMQ, FastAPI) at 99.5% uptime with zero security incidents - the engineering rigor that makes AI systems trustworthy in production, not just impressive in a demo. I use Claude Code as my primary development mode: orchestrating parallel engineering tasks, review cycles, and deployment end-to-end through it. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & Agentic Systems: Python (9yr), LangGraph, LangChain, Multi-Agent Orchestration, RAG Pipelines, ChromaDB, Vector Databases, Embeddings, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Prompt Engineering, HITL, Guardrails, Anthropic API, OpenAI API, Ollama, MCP, OCR, Document Intelligence, Claude Code

Backend & Infrastructure: FastAPI, REST APIs, AWS (S3, EC2, RDS, Glue), Docker, PostgreSQL, MySQL, Oracle, Redis, RabbitMQ, GitHub Actions CI/CD, Microservices, Performance Engineering, Linux, Git, Bash

ML & Data: PyTorch, scikit-learn, XGBoost, Random Forest, Logistic Regression, Isolation Forest, Feature Engineering, NLP, BERT, spaCy, Pandas, NumPy, Tableau, Power BI, TypeScript, React.js, SQL, PL/SQL

PROJECTS

AgentProbe: Open-Source Multi-Provider LLM Evaluation Platform

Production AI · Open Source · LLM Evaluation · Agentic

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions CI/CD · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation platform built end-to-end: multi-provider routing across Anthropic, OpenAI, and Ollama via a unified API abstraction; 15-endpoint FastAPI service; 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate); 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix); CI-integrated regression gate that blocks quality degradation before it reaches users. Zero regressions across 1,240+ test cases. Open-source, deployable in enterprise environments, runs 100% locally at zero cost.

Leasel: 8-Agent Production AI Platform with RAG & HITL Escalation

Production AI · LangGraph · RAG · End-to-End

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface production AI platform built solo from concept to deployment - document intelligence (OCR ingestion, section-aware RAG, 100% citation correctness, 0% unsupported-claim rate), 8-agent LangGraph investigation pipeline (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → Human Escalation) with conditional HITL when confidence is below threshold, and a tenant self-service agent with hard guardrails that never auto-resolves regulated decisions. 93.6% precision, 87.9% F1 across 1,240 test cases on 5 US states + federal regulations. Core rule enforced in code: LLM reasons, human or deterministic logic decides every final compliance call.

AI on FHIR: HIPAA-Compliant Healthcare AI - Full-Stack Production

Healthcare APIs · HIPAA · NLP · Full-Stack

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI system integrating with healthcare FHIR APIs via NLP (BERT embeddings, LLM inference) - translates natural-language queries (EN/ES) into structured FHIR API calls; HIPAA-compliant RBAC with authentication, authorization, and full data isolation; ~60% reduction in manual clinical retrieval time. Demonstrates compliance-sensitive production AI with real external API integration.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production AI Deployment

- Led end-to-end deployment of a production agentic AI system (LangGraph, FastAPI, Python) for an external customer - from technical discovery through production handoff, owning architecture, integration, testing, and delivery solo in weeks. RAG layer grounded every compliance output in cited source regulation; served as sole technical owner through customer handoff.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML prediction pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets across 190+ countries - $R^2 = 0.977$; applied Isolation Forest anomaly detection across 3 research pipelines; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and system design.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned the HUD income-limit data pipeline for 7 years** - federal data ingestion → 180,000 domain records powering eligibility and rent calculations for 100K+ housing units; annual releases with baseline validation, reconciliation, and rollback preparation; zero compliance violations across the ownership period.
- **Enterprise-grade production infrastructure** - AWS S3/EC2/RDS/Glue; Docker (50% scalability improvement, 30% deployment time reduction); RabbitMQ async processing with retry logic and idempotency guarantees; Redis caching; 99.5% uptime across multi-tenant SaaS; 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents across 7yr.
- **Scale and delivery** - 200+ SQL query optimizations (30% performance gain); 20+ full-stack production modules shipped across 120+ Agile sprints; debugging tools reducing triage time 40%; mentored 5+ engineers; 300+ peer code reviews; leveraged Claude Code to accelerate engineering delivery.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; data pipeline for 1.6M housing records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017